

Plasma beta-carotene is not a suitable biomarker of fruit and vegetable intake in german subjects with a long-term high consumption of fruits and v...

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BACKGROUND/OBJECTIVE: beta-Carotene is often used as a marker for the amount of fruit and vegetables consumed, but little is known about plasma beta-carotene concentrations in subjects whose habitual (long-term) diets are characterized by different amounts of foods of plant origin. We compared dietary beta-carotene intake and plasma concentrations in women on habitual diets differing in the consumed amounts of foods of plant origin. METHODS: A comparison of dietary beta-carotene intakes and plasma beta-carotene concentrations in women adhering to an average Western diet (n = 172), wholesome nutrition (following preventive recommendations) (n = 238) or a raw food diet (n = 104). RESULTS: Dietary beta-carotene intake was 5.5, 9.3, 14.7 mg/day for women adhering to an average Western diet, wholesome nutrition and raw food diet, respectively (p < 0.001). Corresponding multivariate adjusted plasma beta-carotene concentrations were 1.07, 1.65, and 1.16 micromol/l, respectively (p < 0.001). Comparable dietary beta-carotene intake resulted in lower multivariate adjusted plasma beta-carotene in women adhering to a raw food diet and average Western diet compared to those on wholesome nutrition (p < 0.001 for all intake groups up to 20 mg/day). The amount of fruit and vegetable intake did not predict plasma beta-carotene levels in women consuming a raw food diet. CONCLUSIONS: Plasma beta-carotene concentrations differed among the diet groups, with highest plasma levels in women adhering to wholesome nutrition. Plasma beta-carotene concentrations may not reflect beta-carotene intake and the amount of fruit and vegetables consumed. 2009 S. Karger AG, Basel.